

An End-to-End AI Agent for Kaggle Competitions

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1.Motivation

- **Challenges of Kaggle competitions**

- **A competition requires the expert loop**

This includes validation design, feature extraction, model selection, and ensemble building. The top entries are separated by a very small margin.

- **Data preparation is repetitive**

The same cleaning, split design, and drift checks are done on every dataset.

- **Why an AI agent**

- **A competition is a *scorable* task**

Every competitor delivers a number, so it can be optimized automatically.

- **A competition requires tireless, systematic search**

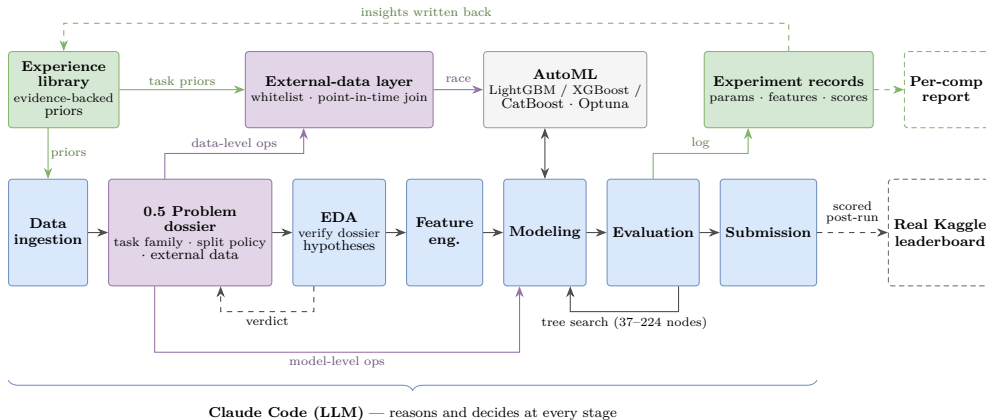
Every experiment is recorded and should be reproducible.

- **Recent success of agents for Kaggle competitions**

Notable cases include tree search over code [1], and tree search fed injected research ideas [2].

2.Workflow

From raw data to a scored submission



The LLM reasons and decides at every stage; the problem dossier classifies the task and injects external data *upstream*, before any modeling. EDA then tests the dossier's hypotheses, feature engineering and modeling follow, and evaluation returns the score the tree search uses to pick the next candidate. Each run also automatically writes a per-competition report.

3.Results

Table 1: ordinary competitions

Competition	Metric	Sinica	NVIDIA	AIDE
s3e1	RMSE ↓	0.55275	0.55550	0.55552
s3e3	AUC ↑	0.87157	0.89537	0.86863
s3e5	QWK ↑	0.57223	0.56974	0.58138
s3e7	AUC ↑	0.90441	0.90458	0.90109
s3e9	RMSE ↓	12.3319	12.2291	12.3067
s3e11	RMSLE ↓	0.29368	0.29624	0.29445
s3e14	MAE ↓	333.186	330.716	332.316
s3e16	MAE ↓	1.33822	1.34001	1.34224
s3e19	SMAPE ↓	49.4088	48.2656	48.3413
s4e11	Acc. ↑	0.94107	0.94076	0.94056
s5e1	MAPE ↓	0.08401	0.14594	0.17917
s5e10	RMSE ↓	0.05576	0.05588	0.05579
s6e1	Error ↓	8.71516	8.73239	8.72191
s6e2	AUC ↑	0.95505	0.95508	0.95516
afsis	MCRMSE ↓	0.48917	0.69855	0.49861
cat-in-the-dat	AUC ↑	0.80243	0.77084	0.80217
conway	MAE ↓	0.10414	0.11189	0.11065
tps-aug-2022	AUC ↑	0.58930	0.58730	0.59106
tps-jan-2022	SMAPE ↓	6.29895	4.63651	10.6664
tps-sep-2022	SMAPE ↓	28.4961	5.48351	9.72095
Best private score		10 / 20	7 / 20	3 / 20

Bold = best of the three; reference agents AIDE [1], NVIDIA [3].

Table 2: special competitions

Competition	Metric	Sinica	NVIDIA	AIDE
us-patent	Pearson r ↑	0.86658	0.86160	0.65821
ventilator	MAE ↓	0.16525	0.40093	0.34591
SIIM-ISIC	AUC ↑	0.93529	0.89822	0.88579

Graded by MLE-bench [4].

4. Disadvantages of the agent-based approach

- **The agent is strong at coding, weak at judging**

The agent efficiently works through whatever problem it is given. However, it does not ask whether a change is worth making. This often leads to a poor design choice that yields bugs.

- **The agent can only target the local validation score**

As a result, the local validation score can look outstanding while the real leaderboard score is mediocre.

- **The agent often buries the main question in detail**

AI-written reports put too much focus on technical details, drifting away from what the reader wants to know.

- **The agent has bad writing habits**

Redundant headings and reinvented terms often come back in every draft. Verifying and modifying them by hand is exhausting.

5. Practical advice for building AI agents for data science

- **Build an isolated environment before running the agent**

This prevents any access to the leaderboard or another run's files.

- **Justify the design before building the pipeline**

This mitigates redesign cost, debugging time, and a full rerun of earlier competitions.

- **Inject external knowledge before data modeling**

This often improves the score. However, if those external data are injected at the final step, the score shows almost no improvement.

6. References

- [1] Z. Jiang *et al.*, AIDE: AI-driven exploration in the space of code, *arXiv:2502.13138*.
- [2] E. Aygün *et al.*, An AI system to help scientists write expert-level empirical software, *Nature* **654**, 909 (2026).
- [3] NVIDIA Corporation, computer code NVIDIA Kaggle Plugin, Santa Clara, CA, 2026, <https://github.com/NVIDIA/nvidia-kaggle>.
- [4] J. S. Chan *et al.*, MLE-bench: Evaluating machine learning agents on machine learning engineering, in *Proceedings of the 13th International Conference on Learning Representations*, 2025, <https://openreview.net/forum?id=6s5uXNWGIh>.

Questions